

## 19.1

What does it mean to assert that the delta of a call option is 0.7? How can a short position in 1,000 options be made delta neutral when the delta of each option is 0.7?

Asserting that the delta of a call option is 0.7 means that when the price of its underlying stock increases or decreases by some amount, the call option's price increases or decreases by 70% of that amount.

To make the short position delta neutral, we see that our short position has delta of  $0.7 \cdot 1000 = 700$ , but as this is a short position, it is -700. So to get to 0, we simply purchase 700 shares.

## 19.6

What is the delta of a short position in 1,000 European call options on silver futures? The options mature in eight months, and the futures contract underlying the option matures in nine months. The current nine-month futures price is \$8 per ounce, the exercise price of the options is \$8, the risk-free interest rate is 12% per annum, and the volatility of silver futures prices is 18% per annum.

## 19.7

In Problem 19.6, what initial position in nine-month silver futures is necessary for delta hedging? If silver itself is used, what is the initial position? If one-year silver futures are used, what is the initial position? Assume no storage costs for silver.

## 19.20

A financial institution has the following portfolio of over-the-counter options on sterling:

| Type | Position | Delta of Option | Gamma of Option | Vega of Option |
|------|----------|-----------------|-----------------|----------------|
| Call | -1,000   | 0.5             | 2.2             | 1.8            |
| Call | -500     | 0.8             | 0.6             | 0.2            |
| Put  | -2,000   | -0.40           | 1.3             | 0.7            |
| Call | -500     | 0.70            | 1.8             | 1.4            |

A traded option is available with a delta of 0.6, a gamma of 1.5, and a vega of 0.8.

(a)

What position in the traded option and in sterling would make the portfolio both gamma neutral and delta neutral?

(b)

What position in the traded option and in sterling would make the portfolio both vega neutral and delta neutral?